OrdinalFed: The Formulas, Explained Simply

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How to read this. Each formula has: **what it does** (one line), the formula, what each **symbol** means, a small **example** with real numbers, and the idea **in one sentence**. You do not need to remember the formulas. If someone asks, explain the "one sentence" and give the example. A one-page cheat sheet is at the end.

0. Symbols you will see everywhere

k	one hospital ("client"). We have $K = 5$ hospitals.
t	the round number: $t = 1, 2, \ldots, T$, with $T = 10$ rounds.
n_k	the number of training images at hospital k.
w	the model weights: all the numbers inside the neural network.
$\tilde{w}_k^t$	hospital k's model after it trained in round t (the "tilde" $\sim$ means "after local training").
y	the true grade of an image (0, 1, 2, 3 or 4).
p_c	the probability the model gives to grade c. All five add up to 1.
$\sum$	"add them all up". Example: $\sum_{c=0}^{4} p_c = p_0 + p_1 + p_2 + p_3 + p_4 = 1$.
e^x **or** $\exp(x)$	the number $e \approx 2.718$ raised to the power x. It is always positive and grows with x.
$\log$	the natural logarithm, the opposite of e^x. Note: $-\log$ of a small number is big.

1. FedAvg: the usual way to combine models

background (Sec. II)

What it does: the server averages the hospitals' models, and bigger hospitals count more.

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{\sum_j n_j} \tilde{w}_k^t$$

The symbols:

- $\frac{n_k}{\sum_j n_j}$ = hospital k's **share of all images** = its vote.
- w^{t+1} = the new global model for the next round.

Example. A city hospital has 600 images, an eye hospital 300, a village clinic 100 (1,000 in total). Votes: $600/1000 = 60\%, 30\%, 10\%$. If one weight inside their models is 2.0, 4.0 and 6.0, the new weight is $0.6(2.0) + 0.3(4.0) + 0.1(6.0) = 1.2 + 1.2 + 0.6 = \mathbf{3.0}$. The city hospital pulls the answer towards its own value.

In one sentence: FedAvg is a weighted average where your vote is your share of the images.

2. Cross-entropy: the usual training loss

Sec. III-B

What it does: measures how wrong the model is, using only the probability it gave to the true grade.

$$\mathrm{CE} \;=\; -\log p_y$$

The symbols:

- p_y = the probability the model gave to the **true** grade y.
- In our paper each grade also has a class weight $\mathbf{c} = [1, 5, 1.5, 10, 4]$, so rare grades count more. We leave it out in these examples to keep the numbers simple.

Example. True grade 4. If the model gives Grade 4 a probability of 0.7: $\mathrm{CE} = -\log 0.7 = 0.36$ (small, good). If 0.2: $-\log 0.2 = 1.61$. If 0.05: $-\log 0.05 = 3.00$ (big, bad). **But:** CE does not care where the rest of the probability went, to Grade 3 (close) or Grade 0 (far).

In one sentence: cross-entropy only asks "how sure were you about the right answer?", not "how far off were you?".

3. Expected grade

Paper Eq. (2)

What it does: turns the five probabilities into one "average" grade.

$$\hat{y} \;=\; \sum_{c=0}^{4} c \cdot p_c \;=\; 0 \cdot p_0 + 1 \cdot p_1 + 2 \cdot p_2 + 3 \cdot p_3 + 4 \cdot p_4$$

Example. The model says: 10% Grade 0, 20% Grade 1, 70% Grade 2. Then $\hat{y} = 0(0.1) + 1(0.2) + 2(0.7) = 0 + 0.2 + 1.4 = \mathbf{1.6}$. So the model's "average guess" is about Grade 1.6.

In one sentence: the expected grade is the model's average guess, using its probabilities as weights.

4. OrdinalLoss: our loss at each hospital

Paper Eq. (1)

What it does: cross-entropy plus a penalty that grows when the guess is far from the true grade.

$$\mathcal{L}_{\text{ord}} \;=\; (1-\lambda)\,\mathrm{CE} \;+\; \lambda\,(\hat{y}-y)^2 \qquad \text{with } \lambda = 0.5$$

The symbols:

- $(\hat{y}-y)^2$ = the **distance penalty**: the gap between the average guess and the true grade, squared (so far misses cost much more).
- λ (lambda) = how much of each part. $\lambda = 0.5$ means half CE, half distance penalty.

Example. True grade 4. Two models both give Grade 4 a probability of 0.2, so **both have the same CE = 1.61**.

Model A puts the rest near Grade 4: $p = [0, 0, 0.1, 0.7, 0.2]$, so $\hat{y} = 0.2 + 2.1 + 0.8 = 3.1$ and the penalty is $(3.1-4)^2 = 0.81$. Loss $= 0.5(1.61) + 0.5(0.81) = \mathbf{1.21}$.

Model B puts the rest on Grade 0: $p = [0.7, 0.1, 0, 0, 0.2]$, so $\hat{y} = 0.1 + 0.8 = 0.9$ and the penalty is $(0.9-4)^2 = 9.61$. Loss $= 0.5(1.61) + 0.5(9.61) = \mathbf{5.61}$.

CE could not tell them apart. OrdinalLoss punishes Model B more than 4 times as much. (The slide's "16" is the extreme case: a sure guess of 0 gives $(0-4)^2 = 16$, a sure guess of 3 gives $(3-4)^2 = 1$.)

In one sentence: OrdinalLoss = cross-entropy + "how far off was your average guess, squared".

5. QWK: our score

Sec. I, used everywhere

What it does: gives one number for how well predicted grades agree with true grades, punishing far mistakes more and correcting for lucky guessing.

$$\kappa \;=\; 1 \;-\; \frac{\sum_{i,j} W_{ij}\, O_{ij}}{\sum_{i,j} W_{ij}\, E_{ij}} \qquad \text{with} \qquad W_{ij} = \frac{(i-j)^2}{(5-1)^2}$$

The symbols:

- O_{ij} = how many images with true grade i were **predicted** as grade j (what really happened).
- E_{ij} = how many we would **expect** if the model guessed at random (with the same totals).
- W_{ij} = the penalty for that mistake. It grows with the **square** of the distance:

Distance between true and predicted grade	0	1	2	3	4
Penalty W	0	0.06	0.25	0.56	1

Example. So $\kappa = 1 - \dfrac{\text{our weighted mistakes}}{\text{a random guesser's weighted mistakes}}$. No mistakes $\Rightarrow \kappa = 1$ (perfect). As bad as random guessing $\Rightarrow \kappa = 0$. If our weighted mistakes are 30% of a random guesser's, $\kappa = 1 - 0.3 = \mathbf{0.7}$. A miss by 4 grades costs 16 times a miss by 1 grade (1 vs. 0.06).

In one sentence: QWK compares your mistakes with a random guesser's, and far mistakes cost much more.

6. The server's exam score for each hospital

Paper Eq. (3)

What it does: the server tests each hospital's model on its 50 images and computes its QWK.

$$\kappa_k^t \;=\; \mathrm{QWK}\big(\text{true grades of the 50 images},\ \text{hospital } k\text{'s predictions on them}\big)$$

Example. In round 3, the eye hospital's model gets QWK 0.80 on the 50 images, the city hospital's 0.50, and the village clinic's 0.05. These three numbers are κ^3_{eye}, κ^3_{city}, $\kappa^3_{\text{village}}$. (Made-up numbers, to show the idea.)

In one sentence: κ_k^t is hospital k's mark on the server's exam in round t.

7. β (beta): how fast we switch from size to score

Paper Eq. (4)

What it does: a number that starts at 0.5 and slowly falls to 0 over the 10 rounds.

$$\beta(t) \;=\; \beta_0 \cos\!\left(\frac{\pi\,(t-1)}{2\,(T-1)}\right) \qquad \text{with } \beta_0 = 0.5,\ T = 10$$

The symbols:

- The part inside $\cos(\cdot)$ goes from $0°$ (round 1) to $90°$ (round 10) in steps of $10°$. Cosine goes from 1 to 0, smoothly.

Round t	1	2	3	4	5	6	7	8	9	10
$\beta(t)$	0.50	0.49	0.47	0.43	0.38	0.32	0.25	0.17	0.09	0.00

In one sentence: β is the share of the vote still decided by size; it shrinks smoothly from half to nothing.

8. Quality weights: turning exam scores into votes

Paper Eq. (5)

What it does: turns the hospitals' exam scores into votes that add up to 1 (100%).

$$q_k^t \;=\; \frac{\exp\big(\max(\kappa_k^t,\ \kappa_{\min})/\tau\big)}{\sum_{j=1}^{K}\exp\big(\max(\kappa_j^t,\ \kappa_{\min})/\tau\big)} \qquad \text{with } \kappa_{\min}=0.1,\ \tau=3$$

The symbols:

- $\max(\kappa, 0.1)$ = the **floor**: any score below 0.1 counts as 0.1, so one bad round cannot wipe a hospital out.
- τ (tau) = the **temperature**. A big τ makes the votes almost equal; a small τ gives the best hospital most of the vote.
- Dividing by the sum makes all votes add up to 1. (This whole formula is called a *softmax*.)

Example. Scores: eye 0.80, city 0.50, village 0.05 $\rightarrow$ after the floor: 0.80, 0.50, **0.10**.
With $\tau = 3$: divide by 3 $\rightarrow$ 0.267, 0.167, 0.033; take e^x $\rightarrow$ 1.306, 1.181, 1.034; sum = 3.521.
Votes: eye $1.306/3.521 = \mathbf{37}\%$, city $\mathbf{34}\%$, village $\mathbf{29}\%$: close to equal.
With $\tau = 0.5$ (sharper): the votes would be 56%, 31% and 14%: the best hospital wins much more.
With $\tau = 3$, the best possible score (1.0) against the floor (0.1) gives at most $e^{(1.0-0.1)/3} = e^{0.3} \approx \mathbf{1.35}$ times the vote. That is why our votes stay close to equal.

In one sentence: a softmax turns exam marks into votes; with $\tau = 3$ it is gentle, so votes stay close to equal.

9. The final vote: mixing size and score

Paper Eq. (6)

What it does: each hospital's final vote is part "share of images" and part "exam vote"; β decides the mix.

$$\omega_k^t \;=\; \beta(t)\,\frac{n_k}{\sum_j n_j} \;+\; \big(1-\beta(t)\big)\,q_k^t$$

The symbols:

- ω_k^t (omega) = hospital k's final vote in round t.
- First part = FedAvg's vote (size). Second part = the exam vote from Formula 8.

Example. Same three hospitals: image shares 60% (city), 30% (eye), 10% (village); exam votes 34%, 37%, 29%.

	City	Eye hospital	Village clinic
FedAvg (size only)	60%	30%	**10%**
OrdinalFed, round 1 ($\beta = 0.5$)	47%	34%	**20%**
OrdinalFed, round 5 ($\beta = 0.38$)	44%	34%	**22%**
OrdinalFed, round 10 ($\beta = 0$)	34%	37%	**29%**

Round 1 for the village: $0.5 \times 10\% + 0.5 \times 29\% = 5\% + 14.7\% \approx 20\%$. The village's vote grows from 10% to 29%. (In our real experiment, the two smallest hospitals grew from 5.9% to 40.8%.)

In one sentence: the final vote starts as half size, half exam score, and slowly becomes all exam score.

10. The safety rule: a minimum vote

Sec. III-F

What it does: every hospital gets at least 3% of the vote, then all votes are rescaled to add up to 1.

$$\omega_k^t \leftarrow \max(\omega_k^t,\ 0.03), \qquad \text{then divide every vote by their sum}$$

Example. In our setting it never activates: the smallest exam vote with 5 hospitals is at least about 16%, and in round 1 it still counts half, so every hospital already has at least about $0.5 \times 16\% = 8\% > 3\%$.

In one sentence: a seat-belt rule: nobody's vote can drop below 3%.

11. The new global model

Sec. III-F, Algorithm 1

What it does: the same weighted average as FedAvg, but with OrdinalFed's votes.

$$w^{t+1} = \sum_{k=1}^{K} \omega_k^t \, \tilde{w}_k^t$$

Example. One weight inside the models is 2.0 (city), 4.0 (eye), 6.0 (village).
FedAvg: $0.60(2) + 0.30(4) + 0.10(6) = \mathbf{3.0}$.
OrdinalFed round 1: $0.47(2) + 0.34(4) + 0.20(6) \approx \mathbf{3.46}$. Round 10: $0.34(2) + 0.37(4) + 0.29(6) \approx \mathbf{3.92}$.
The smaller hospitals now pull the global model too.

In one sentence: same averaging as FedAvg, but with fairer votes.

12. FedProx's "rubber band"

baseline, Sec. IV-C

What it does: each hospital adds a penalty for moving its model too far from the global model.

$$\mathcal{L}_{\text{FedProx}} = \mathcal{L} + \frac{\mu}{2} \left\| w - w^t \right\|^2 \qquad \text{with } \mu = 0.01$$

The symbols:

- $\|w - w^t\|^2$ = for every weight, take (local value $-$ global value)^2, and add them all up.
- μ (mu) = how strong the rubber band is.

Example. The global value of one weight is 1.0. A hospital moves it to 3.0. That weight's penalty is $\frac{0.01}{2}(3.0 - 1.0)^2 = 0.005 \times 4 = 0.02$. It is small for one weight, but the model has millions of weights, so together the penalty keeps the hospital close to the global model.

In one sentence: FedProx ties each hospital's model to the global one with a rubber band; this needs a change in every hospital's training.

13. Recall (per grade)

Table IV

What it does: of all the images that truly are a grade, the share the model found.

$$\text{Recall} \;=\; \frac{\text{Grade 3 images the model called Grade 3}}{\text{all true Grade 3 images}}$$

Example. There are 100 true Grade 3 images. OrdinalFed finds 36 of them: recall $= 36/100 = \mathbf{0.36}$. FedProx finds 31: recall $= 0.31$.

In one sentence: recall answers “out of all the real cases, how many did we catch?”.

One-page cheat sheet

Formula	Name	In plain words
$w^{t+1} = \sum_k \frac{n_k}{\sum_j n_j} \tilde{w}_k^t$	FedAvg	Average the models; vote = share of images.
$\mathrm{CE} = -\log p_y$	Cross-entropy	"How sure were you about the right answer?"
$\hat{y} = \sum_c c\, p_c$	Expected grade (Eq. 2)	The model's average guess.
$(1-\lambda)\mathrm{CE} + \lambda(\hat{y}-y)^2$	OrdinalLoss (Eq. 1)	CE + "how far off, squared". $\lambda = 0.5$.
$1 - \frac{\sum WO}{\sum WE}, W = \frac{(i-j)^2}{16}$	QWK	Your mistakes vs. a random guesser's; far misses cost more.
$\kappa_k^t = \mathrm{QWK}(\ldots)$	Exam score (Eq. 3)	Each hospital's mark on the 50 server images.
$\beta_0 \cos\left(\frac{\pi(t-1)}{2(T-1)}\right)$	β schedule (Eq. 4)	Size counts less each round: 0.5 $\to$ 0.
$\frac{e^{\max(\kappa_k,\, 0.1)/\tau}}{\sum_j e^{\max(\kappa_j,\, 0.1)/\tau}}$	Quality vote (Eq. 5)	Marks $\to$ votes; $\tau = 3$ keeps them gentle (max 1.35$\times$).
$\beta \frac{n_k}{\sum n} + (1-\beta) q_k$	Final vote (Eq. 6)	Half size, half score at first; all score at the end.
$\max(\omega_k, 0.03)$	Minimum vote	Seat belt: nobody below 3%.
$\sum_k \omega_k \tilde{w}_k^t$	New global model	Same averaging, fairer votes.
$\mathcal{L} + \frac{\mu}{2}\lVert w - w^t \rVert^2$	FedProx	Rubber band to the global model; $\mu = 0.01$.
$\frac{\text{found}}{\text{all true}}$	Recall	How many real cases we caught.

If someone asks about a formula, say it in three steps:
1. What it is for ("this sets each hospital's vote"). 2. The one sentence ("half size, half exam score at first, all exam score at the end"). 3. One number from the example ("the village goes from 10% to 29%").